

Hybrid Model-Data-Driven Viscous Loss Modeling and Analysis for Oil-Flooded High-Speed Electrical Machines in Electric Actuators

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With the trend toward electrification in the aerospace sector, oil-flooded high-speed permanent-magnet (PM) machines are gaining increasing attention for electric actuator applications due to their high power density and compactness. However, as operational speeds rise, viscous losses become dominant and critically affect machine performance and reliability, yet accurate prediction remains challenging due to complex flow regimes and limited high-fidelity data. This

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article presents a hybrid mode-data-driven approach for temperature-dependent viscous loss modeling of oil-flooded high-speed PM machines with coupled temperature-loss analysis. A transfer-learning-based neural network is developed to construct an accurate surrogate model by leveraging low-cost analytical data and limited high-fidelity computational fluid dynamics samples. The proposed model captures the temperature-dependent viscous loss behavior and is further embedded into a temperature-aware performance evaluation framework in which oil temperature and viscous loss are updated iteratively. Finally, an oil-flooded high-speed PM machine is built to validate the proposed approach, and experimental results across a wide operating range confirm the accuracy and effectiveness of the proposed approach.

I. INTRODUCTION

The electrification of aerospace sector is progressing rapidly, with electro-hydrostatic actuators (EHAs) emerging as one of the key enablers for power-by-wire flight control systems in more-electric aircraft [1], [2]. EHAs can be regarded as an intermediate solution between centralized hydraulic actuation and fully electric actuation, as they combine local electric drive with hydraulic power transmission [2]. As a result, EHAs offer high energy efficiency, high reliability, and ease of maintenance [3], [4], and have already been successfully implemented in advanced platforms, such as the Airbus A380 [2]. As the core power conversion component of the EHA, motor-driven pumps are increasingly designed in an integrated manner to meet stringent requirements on reliability, size, and system compactness [5], [6]. In this context, oil-flooded high-speed permanent-magnet (PM) machines are receiving considerable attention due to their high level of integration, superior thermal management, and the removal of shaft seals, which contribute significantly to improved reliability and compactness of the overall system.

However, the oil-flooded configuration introduces hydraulic oil directly into the airgap, leading to considerable viscous losses. With the increasing demand for higher power density (kW/kg or kW/L), the maximum rotational speed of such machines has reached 20 000 r/min [6]. Under such high-speed conditions, viscous loss becomes a major source of internal heat generation [7]. Therefore, fast and accurate prediction of viscous loss is crucial for thermal modeling, which underpins machine design optimization and reliability evaluation.

Existing approaches for estimating viscous loss can be broadly classified into numerical and analytical methods. Numerical method, primarily computational fluid dynamics (CFD), offers detailed insight into fluid behavior and provides accurate viscous loss predictions [8], but its high computational cost, strict meshing requirements, and difficulty in parameterization restrict its applicability to case-specific analyses. Analytical methods, by contrast, offer faster computation and are widely used in machine design practice. In laminar flow conditions, analytical expressions can be directly derived. Under turbulent flow, empirical formulas are often developed by fitting experimental [9], [10] or CFD [11] data, and are also considered part of the

analytical category in this article. Several studies have applied such models to estimate viscous losses in oil-flooded high-speed PM machines for aerospace EHA. Nonetheless, the accuracy and generalizability of these analytical models are often limited by the machine geometry and operating conditions used for fitting [12], which hinders their broader applicability in design optimization.

In recent years, machine learning techniques have provided data-driven solutions for modeling [13] and design optimization [14], [15] of electric machines. However, the accuracy and generalization capability of surrogate models are highly dependent on the size, quality, and coverage of the training dataset, and limited data often lead to poor generalization [16]. Transfer learning (TL) has shown significant potential in addressing this challenge by leveraging prior knowledge from source domains, such as analytical datasets, to enable accurate and data-efficient surrogate modeling with reduced reliance on high-fidelity data. In [17], TL-based framework has been used to build electromagnetic surrogate models for five-phase machine optimization by transferring knowledge from analytical to finite element domains. A multifidelity surrogate model is proposed for multidisciplinary rotor optimization in high-speed PM machines [18], following a similar concept. In [19], a modular neural network approach based on TL has also been proposed to mitigate the need for retraining surrogate models under varying design parameters in multiobjective optimization of interior PM machines. These studies highlight the effectiveness of surrogate modeling using TL and provide a methodological reference for viscous loss prediction with limited data in oil-flooded high-speed PM machines.

In oil-flooded high-speed machines, viscous losses are strongly temperature dependent, as fluid viscosity and flow regimes vary significantly with operating temperature. This leads to nonlinear coupling between loss generation and temperature evolution, making accurate prediction challenging. Meanwhile, lumped-parameter thermal networks (LPTNs) have been widely adopted for thermal analysis of electrical machines due to their low computational cost and structural flexibility [20], [21], [22], [23]. In [21], an LPTN model is developed for oil-flooded machines with slot-channel cooling, achieving temperature prediction errors within 3%. However, accurate prediction of temperature-dependent viscous losses requires resolving the nonlinear coupling between loss and temperature, which generally necessitates fine time-step numerical computation (e.g., 0.0001 s) [21] and is computationally intensive [23]. This highlights the need for efficient modeling approaches that can accurately capture temperature-dependent viscous losses while enabling fast coupled temperature-loss evaluation.

To address the challenges of accurate and computationally efficient temperature-dependent viscous loss prediction in oil-flooded high-speed machines, this article proposes a hybrid model-data-driven (HMDD) approach that combines a TL-based viscous loss surrogate model with an analytically solvable LPTN model for performance evaluation.

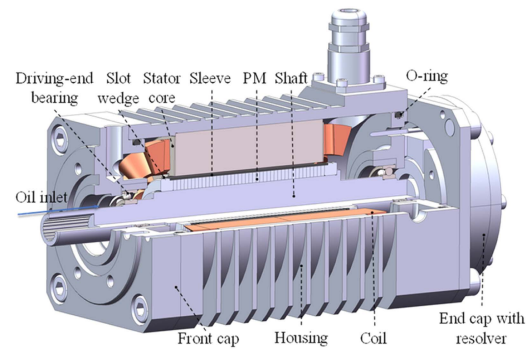


Fig. 1. Basic structure of the oil-flooded high-speed PM machine.

Comprehensive experimental validation is conducted to assess the accuracy and applicability of the proposed method over a wide operating range. The main contributions of this work are summarized as follows.

- 1) A TL-based viscous loss surrogate model is developed by transferring prior knowledge from low-cost analytical data and fine-tuning with limited high-fidelity CFD samples, enabling accurate prediction over a wide speed–temperature range. The impact of transfer dataset size on model performance is also evaluated to highlight its effectiveness under limited high-fidelity data.
- 2) An HMDD framework for temperature-dependent viscous loss analysis in oil-flooded high-speed machines is developed by coupling the learned viscous loss model with a temperature-aware model using an analytically derived solution, enabling rapid and accurate prediction of temperature-loss interaction under transient operating conditions.
- 3) Experimental validation over broad speed (3000–27000 r/min) and temperature (20 °C–100 °C) ranges confirms the accuracy and applicability of the proposed HMDD approach, offering a modeling foundation and experimental reference for performance evaluation and design optimization of oil-flooded high-speed machines in EHA applications.

The rest of this article is organized as follows. Section II develops both analytical and CFD-based models for viscous loss and proposes a TL-based surrogate modeling method. Section III introduces a temperature-aware model with an analytically derived solution, outlining the overall HMDD viscous loss analysis framework. Section IV evaluates the performance of the surrogate model and validates the HMDD approach through experimental results. Finally, Section V concludes this article.

II. TL-BASED VISCOUS LOSS MODELING

Fig. 1 illustrates the typical structure of the oil-flooded high-speed PM machine. In the EHA configuration considered in this work, leakage oil from the axial piston pump enters the machine through the driving-end bearing and fills the internal cavity, while no forced inlet–outlet axial oil flow

is introduced through the airgap during operation or viscous loss testing. As a result, the viscous losses induced by the oil-flooded configuration and the resulting temperature rise require careful consideration. Slot wedges are adopted to seal the stator slot openings and form a closed-slot configuration, thereby reducing slot opening disturbances and simplifying the airgap flow model.

A. Analytical Model

Considering that the oil-flooded machine adopts the closed-slot structure and no forced axial oil flow is introduced through the air gap, the disturbances caused by stator slot openings and axial flow components are neglected in the model. Therefore, the oil flow in the machine airgap can be reasonably approximated by the Taylor–Couette (TC) flow model, which describes fluid behavior in the annular space between two coaxial rotating cylinders. For laminar conditions (typically for Reynolds number $Re_\delta < 300$), the tangential velocity u exhibits a linear radial distribution, i.e., the velocity gradient du/dr remains constant. Based on Newton’s law of viscosity, the shear stress τ_L on the rotor surface can be analytically derived as

$$\tau_L = \mu r \frac{\partial}{\partial r} \left(\frac{u}{r} \right) = -2\mu\omega_r \frac{R_{si}^2}{R_{si}^2 - R_{ro}^2} \quad (1)$$

where μ is the dynamic viscosity of the fluid, ω_r is the rotor angular velocity, and R_{ro} and R_{si} denote the outer radius of the rotor and the inner radius of the stator, respectively. The corresponding viscous loss $P_V^{(L)}$ is

$$P_V^{(L)} = 4\mu\pi\omega_r^2 l_e \frac{(R_{ro} + \delta)^2 R_{ro}^2}{\delta (2R_{ro} + \delta)}, \quad Re_\delta \leq 300 \quad (2)$$

$$Re_\delta = \frac{\rho u_p \delta}{\mu} \quad (3)$$

where l_e is the active length of the machine, δ is the airgap width, $u_p = R_{ro}\omega_r$ is the peripheral velocity of the rotor, and ρ is the fluid density.

As Re_δ increases, inertial effects become more prominent, enabling the fluid to overcome viscous constraints and develop radial secondary flows, leading to the formation of steady Taylor vortices, i.e., the transition regime. With further increase in Re_δ , the flow evolves into a fully developed turbulent state. In such transitional and turbulent regimes, the shear stress τ_T is expressed as [24]

$$\tau_T = (\mu + \eta) r \left(\frac{d\bar{u}}{dr} \right) \quad (4)$$

where $\bar{u}$ is the time-averaged tangential velocity and η is the eddy viscosity. Due to its spatial nonuniformity and dependence on local flow state, η exhibits strong nonlinearity and is generally difficult to resolve analytically. Therefore, empirical models validated by dimensionless analysis are typically employed to estimate viscous losses under transitional and turbulent flow regimes, expressed as [10]

$$P_V^{(T)} = C_f \rho \pi \omega_r^3 R_{ro}^4 l_e \quad (5)$$

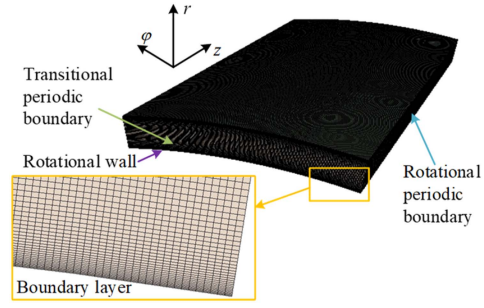


Fig. 2. Mesh and boundary condition of the oil flow in the airgap.

$$\begin{cases} C_f = 0.515 \frac{(\delta/R_{ro})^{0.3}}{Re_\delta^{0.5}}, & 300 < Re_\delta < 10\,000 \\ C_f = 0.0325 \frac{(\delta/R_{ro})^{0.3}}{Re_\delta^{0.2}}, & Re_\delta \geq 10\,000. \end{cases} \quad (6)$$

B. Numerical Model

Given the limitations of analytical models in transitional and turbulent regimes, a CFD-based numerical model of the airgap flow field is developed for accurate prediction of flow characteristics and viscous losses. The Navier–Stokes equations are time-averaged using Reynolds decomposition, and the governing equations are transformed into the Reynolds-Averaged Navier–Stokes (RANS) equations, given by [22]

$$\begin{cases} \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i} (\rho \bar{u}_i) = 0 \\ \frac{\partial}{\partial t} (\rho \bar{u}_i) + \frac{\partial}{\partial x_j} (\rho \bar{u}_j \bar{u}_i) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial \bar{u}_i}{\partial x_j} - \rho \overline{u'_i u'_j} \right) \end{cases} \quad (7)$$

where p is the pressure and $\bar{u}$ and u' denote the time-averaged component and fluctuating component, respectively. The term $-\rho \overline{u'_i u'_j}$ is referred to as the nonlinear Reynolds stress tensor, which introduces a closure problem in RANS equations for solving, leading to various turbulence models.

In this study, the shear stress transport (SST) k - ω turbulence model is employed to characterize the TC flow and associated viscous losses within the narrow airgap [11]. By combining the near-wall resolution of the k - ω model with the robustness of the k - ε formulation in fully developed regions, the SST k - ω model enables accurate prediction of both strong near-wall shear layers and core flow features [22], [25], making it well-suited for modeling wall-constrained TC flow in oil-flooded machines.

To reduce computational cost, a periodic domain model is constructed, with mesh and boundary settings illustrated in Fig. 2. Mesh quality and density critically influence both the accuracy and efficiency of the numerical solution. Since the SST k - ω model directly resolves the viscous sublayer, high-quality boundary layer meshing is essential. The dimensionless wall distance y^+ is adopted to assess near-wall mesh quality, defined as

$$y^+ = \frac{y \rho u_\tau}{\mu}, \quad u_\tau = \sqrt{\frac{\tau_\omega}{\rho}} \quad (8)$$

where y is the normal distance from the wall, u_τ is the friction velocity, and τ_ω is the wall shear stress. To ensure accurate resolution of the viscous sublayer, the first mesh

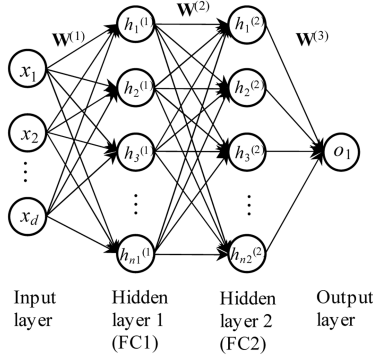


Fig. 3. Architecture of the FNN with two hidden layers.

layer in the radial direction is set to achieve $y^+ \approx 1$, and a mesh growth rate of 1.05 is applied. Mesh independence verification is performed under periodic boundary conditions in both axial and circumferential directions, with a torque variation threshold of less than 0.5% serving as the convergence criterion.

C. TL-Based Surrogate Modeling

1) *Framework*: Given a source domain $\mathcal{D}_s$ with task $\mathcal{T}_s$ and a target domain $\mathcal{D}_t$ with task $\mathcal{T}_t$, TL aims to enhance the performance of a predictive function $f(\cdot)$ in the target domain by leveraging latent knowledge learned from the source domain and task, enhancing learning efficiency or reducing the need for labeled data [26]. In deep TL, $f(\cdot)$ is typically implemented as a deep neural network.

For viscous loss prediction, the analytical model (see Section II-A) and the CFD-based numerical model (see Section II-B) exhibit structural similarities but differ in fidelity, which reflects the level of physical detail and prediction accuracy. The analytical model, with its closed-form mathematical expressions and low computational cost, can generate abundant labeled data and thus serves as an effective source domain for pretraining. Although its accuracy is limited under high-speed and complex-flow conditions, it captures the global variation trend of viscous losses and provides useful prior knowledge. In contrast, the CFD model, while offering high prediction accuracy, is computationally intensive and is therefore adopted as the target domain, where model fine-tuning is performed using a limited set of high-fidelity samples. Building on this complementary relationship, a TL-based viscous loss prediction framework is proposed to correct the bias of the analytical model and develop a high-accuracy surrogate model that leverages both physics-based modeling and data-driven learning. Once trained, the TL-feedforward neural network (FNN) surrogate replaces repeated CFD simulations and enables rapid viscous loss prediction over the speed-temperature operating range, thereby reducing computational cost while preserving the accuracy advantage of high-fidelity CFD samples.

2) *Feedforward Neural Network*: To implement this framework, an FNN with two hidden layers is constructed, as illustrated in Fig. 3. The network architecture consists

of an input layer, two fully connected hidden layers, and an output layer. Through a series of composite nonlinear transformations, the FNN maps the input $\mathbf{X}$ to the output space $\mathbf{O}$, and is capable of approximating arbitrary input-output mappings with desired accuracy [27].

At the input layer ($l = 0$), the input features are normalized to ensure numerical stability. The output vector of each neuron at the input layer, denoted as $\mathbf{H}^{(0)}$, is computed as

$$\mathbf{H}^{(0)} = \text{normalization}(\mathbf{X}). \quad (9)$$

For neurons in hidden layers ($l = 1, \dots, L-1$), the output $\mathbf{H}^{(l)}$ is calculated by applying a nonlinear activation function $\sigma(\cdot)$ to the weighted sum of inputs from the previous layer $\mathbf{H}^{(l-1)}\mathbf{W}^{(l)}$ and an associated bias $\mathbf{b}^{(l)}$

$$\mathbf{H}^{(l)} = \sigma^{(l)}(\mathbf{H}^{(l-1)}\mathbf{W}^{(l)} + \mathbf{b}^{(l)}). \quad (10)$$

The final output of the layer L is expressed as

$$\mathbf{O} = \mathbf{H}^{(L-1)}\mathbf{W}^{(L)} + \mathbf{b}^{(L)}. \quad (11)$$

Consider a source domain training set $\mathcal{D}_s = \{\mathbf{x}_i^{(s)}, y_i^{(s)}\}_{i=1}^{n_s}$, where n_s is the number of training samples, $\mathbf{x}_i^{(s)} = [x_{i1}^{(s)}, x_{i2}^{(s)}, \dots, x_{id}^{(s)}]^T$ represents the d -dimensional input features, and $y_i^{(s)}$ denotes the corresponding target values. The optimization function of the FNN is formulated as

$$\arg \min_{\mathbf{W}, \mathbf{b}} L(\mathbf{W}^{(l)}, \mathbf{b}^{(l)}) + \frac{\lambda_*}{2} \sum_1^L \|\mathbf{W}^{(l)}\|_F^2 \quad (12)$$

where the mean-squared error is used as the loss function for this regression task, given by

$$L(\mathbf{W}^{(l)}, \mathbf{b}^{(l)}) = \frac{1}{n_s} \sum_{i=1}^{n_s} \frac{1}{2} (\hat{y}_i^{(s)} - y_i^{(s)})^2. \quad (13)$$

Furthermore, an L_2 -regularization term is incorporated to improve generalization.

The FNN is optimized using the mini-batch stochastic gradient descent algorithm, with parameters iteratively updated along the negative gradient direction according to

$$\begin{aligned} \mathbf{W}^{(l)} &\leftarrow (1 - \lambda\eta_*)\mathbf{W}^{(l)} - \frac{\eta_*}{|\mathcal{B}|} \sum_{\mathcal{B}} \partial_{\mathbf{W}^{(l)}} L(\mathbf{W}^{(l)}, \mathbf{b}^{(l)}) \\ \mathbf{b}^{(l)} &\leftarrow \mathbf{b}^{(l)} - \frac{\eta_*}{|\mathcal{B}|} \sum_{\mathcal{B}} \partial_{\mathbf{b}^{(l)}} L(\mathbf{W}^{(l)}, \mathbf{b}^{(l)}) \end{aligned} \quad (14)$$

where η_* denotes the learning rate, $|\mathcal{B}|$ denotes the batch size, and λ is the L_2 -regularization coefficient.

3) *Model-Based TL*: To implement TL, a two-stage “pretraining-fine-tuning” strategy is adopted [19]. The FNN is first pretrained on the source domain dataset to capture generalizable features. The pretrained FNN, along with its parameters, is transferred and fine-tuned using a small set of labeled target domain samples, as illustrated in Fig. 4. Following established findings that lower layer features tend to be more transferable while higher layer representations are task-specific [28], the second fully connected layer (FC2) and the output layer are selected as adaptation layers for fine-tuning. This approach enables preserving the learned

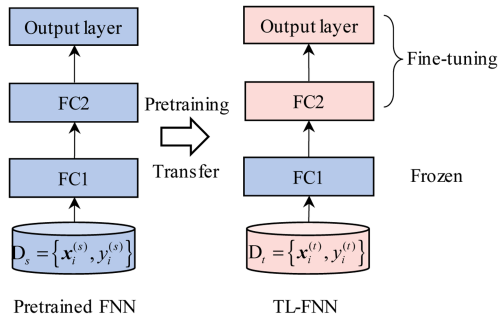


Fig. 4. “Pretraining-fine-tuning” process for the TL-FNN.

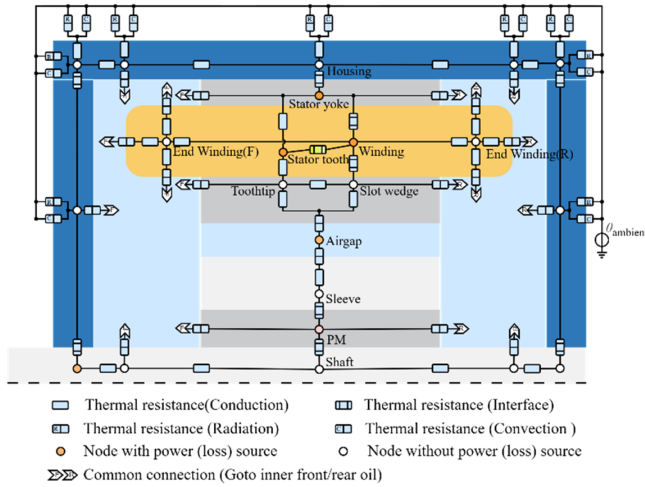


Fig. 5. 3-D LPTN for oil-flooded high-speed PM machines.

source domain knowledge while effectively capturing target domain characteristics. Given the target domain training set $\mathcal{D}_t = \{x_i^{(t)}, y_i^{(t)}\}_{i=1}^{n_t}$, the first fully connected layer (FC1) is frozen (learning rate set to zero) during fine-tuning, while the remaining network parameters are updated with a reduced learning rate based on the training procedure described in (12)–(14). This TL strategy enables the model to adapt effectively to the target domain and achieve accurate predictions.

III. HMDD TEMPERATURE-DEPENDENT VISCOUS LOSS ANALYSIS METHOD

A. Temperature-Aware Modeling

As shown in Fig. 1, in this type of machine, the internal structure is completely flooded in oil, resulting in the airgap and end-winding region being cooled by oil. Therefore, when establishing the thermal model of such machines, in addition to considering conduction between solid components and natural convection and radiation heat transfer at the housing, it is also necessary to account for forced convection in the airgap and end-winding regions, respectively.

A 3-D LPTN is developed for oil-flooded high-speed PM machines with full consideration of both radial and axial heat transfer within the machine, as illustrated in Fig. 5. The losses are assumed to be uniformly distributed within each

equivalent region and are assigned to the corresponding thermal nodes according to their locations. Considering the inherent rotational symmetry of the machine geometry, the circumferential temperature distribution is considered periodic in the LPTN. Accordingly, the machine can be symmetrically divided along the stator tooth and slot centerlines into $2N_s$ (N_s denotes the number of slots) equal segments, allowing thermal analysis to be conducted on a representative half-slot model.

The LPTN is constructed by discretizing structural components into elementary thermal units, each associated with a central node coupled to a corresponding heat source (i.e., machine losses) Φ_s and thermal capacitance C_s . These elementary units consist of cylindrical segments (coaxial hollow cylinders) and cubic elements, selected based on the geometric structure of each component. Node interconnections are established according to physical contact and heat transfer paths. The number of nodes within each region can be adaptively adjusted to balance modeling accuracy and computational efficiency, without altering the network topology or solution method.

For conduction heat transfer within solids, the temperature rise caused by the boundary temperature difference can be described using Fourier’s law of heat conduction. Based on this principle, the conduction thermal resistance of the elementary units can be defined as follows:

$$R_i = \frac{\Delta T_i}{\Phi_i} = \frac{l_i}{\lambda_m A_{\text{cond}}} \quad (15)$$

where R_i , ΔT_i , and Φ_i represent the conduction thermal resistance, temperature difference, and heat flux of the i th module, respectively. λ_m denotes the thermal conductivity of the material and A_{cond} and l_i represent the heat conduction area and the axial length, respectively.

Moreover, due to factors, such as surface roughness and assembling, a small interface gap exists between solid contact surfaces (e.g., between the stator core and the housing), where the heat transfer mechanism is modeled as pure conduction. The thermal resistance of this gap, known as thermal contact resistance R_g , can be calculated as follows:

$$R_g = \frac{\delta_g}{\lambda_g \cdot A_g} \quad (16)$$

where δ_g represents the length of the interface gap, which typically depends on the material and processing of the contact surfaces. λ_g is the thermal conductivity of the interface material, and A_g denotes the contact area. For simplicity, the series-connected conduction thermal resistance and contact thermal resistance are combined into a single thermal resistance (Interface) in Fig. 5.

In terms of convective heat transfer, the convective thermal resistance R_c is calculated by

$$R_c = \frac{1}{h_c \cdot A_{\text{conv}}} \quad (17)$$

where A_{conv} denotes the convection heat exchange area between the fluid and the solid, and the convection heat

transfer coefficient (HTC) h_c is calculated using

$$h_c = \frac{\lambda_f \text{Nu}}{L_c} \quad (18)$$

where λ_f denotes the thermal conductivity of the fluid, L_c is the characteristic length, and Nu is dimensionless Nusselt number, which characterizes the enhancement of heat transfer through a fluid layer by convection relative to conduction across the same fluid layer [29]. Therefore, when calculating the convective HTC in each region, particular attention should be paid to determining the Nusselt number Nu.

For radiative heat transfer, only the radiation from the external surface of the housing is considered. The radiation thermal resistance R_{rad} is calculated by

$$R_{\text{rad}} = \frac{1}{h_r \cdot A_{\text{rad}}} \quad (19)$$

where A_{rad} is the radiative surface area and h_r is the radiation HTC between the housing and environment, given by

$$h_r = \varepsilon \cdot \sigma \cdot (\vartheta_1^3 + \vartheta_1^2 \vartheta_0 + \vartheta_1 \vartheta_0^2 + \vartheta_0^3) \quad (20)$$

where ε is the surface emissivity of the solid, σ is the Stefan-Boltzmann constant, $\sigma = 5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \cdot \text{K}^4)$, and ϑ_1 and ϑ_0 are the surface temperature and ambient temperature, respectively.

The thermal capacitance C_i , defining the amount of heat required to raise a component's temperature by one unit, is calculated as

$$C_i = \rho_{m,i} \cdot V_{m,i} \cdot c_{p,i} \quad (21)$$

where $\rho_{m,i}$ is the material density, $V_{m,i}$ is the component volume, and $c_{p,i}$ is the specific heat.

For the heat sources, the machine losses considered in this article include: 1) stator copper loss, 2) iron loss, 3) PM eddy current (EC) loss, and 4) viscous loss. The stator copper loss P_{Cu} is divided into active winding loss $P_{\text{Cu,AW}}$ and end-winding loss $P_{\text{Cu,EW}}$, calculated as

$$P_{\text{Cu}} = 3I_{\text{RMS}}^2 R_{\text{ph}} (\vartheta_{\text{wdg}}) \quad (22)$$

$$R_{\text{ph}} (\vartheta_{\text{wdg}}) = R_{\text{ph0}} (1 + \alpha_{\text{Cu}} (\vartheta_{\text{wdg}} - \vartheta_{\text{ref}})) \quad (23)$$

$$P_{\text{Cu,AW}} = \frac{l_{\text{AW}}}{l_{\text{AW}} + l_{\text{EW}}} P_{\text{Cu}}, P_{\text{Cu,EW}} = \frac{l_{\text{EW}}}{l_{\text{AW}} + l_{\text{EW}}} P_{\text{Cu}} \quad (24)$$

where l_{AW} and l_{EW} are the lengths of the active and end windings, ϑ_{wdg} and ϑ_{ref} are the winding and reference temperatures, respectively, and R_{ph} and R_{ph0} are the phase resistance at ϑ_{wdg} and ϑ_{ref} , respectively. I_{RMS} is the RMS value of phase current, and $\alpha_{\text{Cu}} = 0.00393$ is the temperature coefficient of copper.

The iron loss P_{Fe} and PM EC loss P_{EC} are directly obtained using electromagnetic finite element analysis method, in which P_{Fe} is calculated using Bertotti's classical formulation

$$P_{\text{Fe}} = C_{\text{hys}} B^2 f + C_{\text{ec}} B^2 f^2 + C_{\text{exc}} B^{1.5} f^{1.5} \quad (25)$$

where B is the magnetic flux density, f is the frequency, and C_{hys} , C_{ec} , and C_{exc} are the coefficients of hysteresis, EC,

and excess iron losses, respectively. The EC loss P_{EC} in the PMs is computed by the volume integration, considering the EC density J_{ec} and the PM electrical conductivity σ_{PM} .

$$P_{\text{EC}} = \int_V \frac{J_{\text{ec}}^2}{\sigma_{\text{PM}}} dV. \quad (26)$$

B. Analytical Solution Formulation

Kirchhoff's law is used to solve the temperature distribution. At steady state, the sum of heat flows entering and leaving each node is zero, and the node temperature equation is expressed as

$$\mathbf{A}_t \Phi_t = \mathbf{0} \quad (27)$$

where $\mathbf{A}_t$ is the incidence matrix describing the connectivity between nodes and branches and Φ_t is the vector of branch heat flux, further expressed as

$$\Phi_t = \mathbf{Y}_t \mathbf{A}_t^T \Theta_{tN} + \mathbf{I}_{tS} \quad (28)$$

where $\mathbf{I}_{tS}$ and Θ_{tN} are the vectors of branch heat flow sources and node temperature, respectively, $\mathbf{Y}_t = \text{diag} \left[\frac{1}{R_1} \frac{1}{R_2} \dots \frac{1}{R_{n_b}} \right]$ is the branch thermal conductance matrix, consisting of the thermal conductance of all n_b branches. Combining (27) and (28), the steady-state node temperatures can be determined as follows:

$$\Theta_{tN} = -(\mathbf{A}_t \mathbf{Y}_t \mathbf{A}_t^T)^{-1} \mathbf{A}_t \mathbf{I}_{tS}. \quad (29)$$

Considering that the viscous loss and stator copper loss, as well as convective and radiative thermal resistances, are temperature-dependent and exhibit nonlinear behavior, an iterative approach is typically required. In each iteration, node temperatures are updated, temperature-dependent parameters are recalculated, and the process is repeated until the node temperatures converge within a defined tolerance.

For transient conditions, the effect of thermal capacitance is introduced, yielding the dynamic form of the node temperature equation, expressed as

$$\mathbf{A}_t \mathbf{Y}_t \mathbf{A}_t^T \Theta_{tN} + \mathbf{A}_t \mathbf{I}_{tS} + \mathbf{C}_t \frac{d\Theta_{tN}}{dt} = \mathbf{0} \quad (30)$$

where $\mathbf{C}_t = \text{diag}[C_1 C_2 \dots C_{n_o}]$ is the matrix of thermal capacitance, C_1 to C_{n_o} represent the thermal capacitance of each node, and n_o denotes the number of nodes.

In transient LPTN analysis, conventional numerical solving methods become increasingly time-consuming as the number of time steps grows. To improve computational efficiency without compromising accuracy, an analytical solution to the transient temperature state equation is developed. Based on the piecewise linearization method, the LPTN is assumed to be quasi-linear within each small time interval, where temperature-dependent thermal parameters and losses are treated as constants, rendering the system time-invariant. After each calculation interval, these temperature-dependent parameters are updated according to the newly calculated temperature, thereby retaining the nonlinear temperature-loss coupling during transient evaluation. Under such assumption, the differential equation

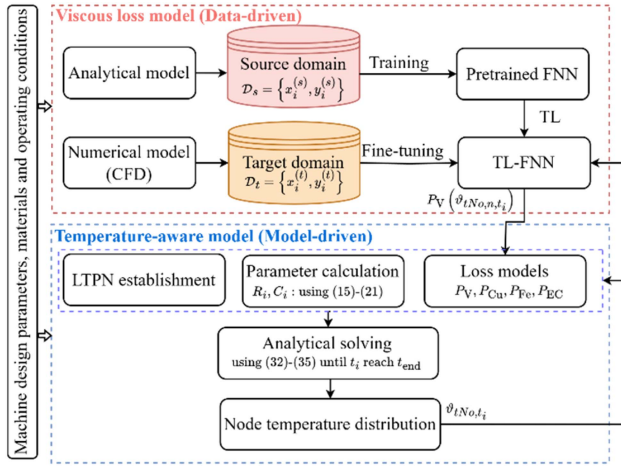


Fig. 6. Overall framework of the proposed HMDD temperature-dependent viscous loss analysis.

in (30) thus admits the analytical solution

$$\vartheta_{tNo}(t) = e^{A_{std}(t-t_0)} \vartheta_{tNo}(t_0) + \int_{t_0}^t e^{A_{std}(t-\tau)} B_{std} I_{tS} d\tau \quad (31)$$

where $A_{std} = -C_t^{-1} A_t Y_t A_t^T$, $B_{std} = -C_t^{-1} A_t$, and t_0 denotes the initial time. Assuming that A_{std} has n_o linearly independent eigenvectors corresponding to n_o distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_{n_o}$, there exists an invertible matrix V that diagonalizes A_{std} via a similarity transformation, given by

$$e^{A_{std}(t)} = V e^{\Lambda(t)} V^{-1}, \Lambda = V^{-1} A_{std} V = \text{diag}(\lambda_1, \lambda_2 \dots \lambda_{n_o}). \quad (32)$$

As a result, the integral in (31) can be solved explicitly, and the transient solution is obtained in closed form, written as

$$\vartheta_{tNo}(t) = V K(t) V^{-1} \vartheta_{tNo}(t_0) + V S(t) V^{-1} B_{std} I_{tS} \quad (33)$$

where

$$K(t) = \text{diag}(e^{\lambda_1(t-t_0)}, \dots, e^{\lambda_{n_o}(t-t_0)})$$

$$S(t) = \text{diag}\left(\frac{1}{\lambda_1} (1 - e^{\lambda_1(t-t_0)}), \dots, \frac{1}{\lambda_{n_o}} (1 - e^{\lambda_{n_o}(t-t_0)})\right). \quad (34)$$

In the discrete time domain, given two consecutive time steps t_i and t_{i+1} , the temperature at t_{i+1} can be calculated using

$$\vartheta_{tNo}(t_{i+1}) = V(\vartheta_{tNo,t_i}) K(\vartheta_{tNo,t_i}, t_i) V^{-1}(\vartheta_{tNo,t_i}) \cdot \vartheta_{tNo,t_i} + V(\vartheta_{tNo,t_i}) S(\vartheta_{tNo,t_i}, t_i) V^{-1}(\vartheta_{tNo,t_i}) B_{std} I_{tS}(\vartheta_{tNo,t_i}) \quad (35)$$

where $\vartheta_{tNo,t_i} = \vartheta_{tNo}(t_i)$.

C. HMDD Viscous Loss Analysis Framework

The overall framework of the proposed HMDD temperature-dependent viscous loss analysis is shown in Fig. 6. First, an analytical model and a CFD-based numerical model are developed based on machine design parameters and operating conditions to generate source and target

TABLE I
Main Specifications and Parameters of the Oil-Flooded High-Speed PM Machine

Parameter	Value	Parameter	Value
Number of phases	3	Number of poles/slots	8/9
Peak speed	22 000r/min	Peak torque	8 N·m
Peak power	18.4 kW	Active-parts weight	2.2 kg
DC bus voltage	270 Vdc	Cooling condition	Oil-flooded (internal); Natural convection (external)
Stator inner diameter	39.24 mm	Rotor diameter	37.60 mm
Motor active length	82 mm	Magnet height	5 mm

TABLE II
Main Material and Thermal Properties of the Oil-Flooded High-Speed PM Machine

Component	Material	Thermal conductivity (W/m/K)	Specific heat (J/kg/K)	Mass density (kg/m ³)
Stator core	ST100	30 ($r-\varphi$) 3.5 (z)	460	7650
Coil	Copper	401	400	8900
	Insulation	0.2	1200	1400
PM (Halbach)	Sm ₂ Co ₁₇	12	350	8400
Rotor sleeve	Carbon fiber	9.3 (φ) 3.5 (r, z)	1000	1600
Housing	Aluminum	230	900	2700
Oil @20°C	15# hydraulic oil	845	0.18	1793

domain datasets for viscous loss prediction. Pretraining and TL are performed then, yielding a TL-FNN-based viscous loss surrogate predictor, which is primarily a data-driven process. Next, the temperature-aware model is constructed from machine geometry and material properties, where heat transfer parameters and multi-source losses are incorporated for performance evaluation in this model-driven stage. Finally, the hybrid coupled system is solved under both steady-state and transient conditions, with iterative updates of temperature and loss parameters, in particular the temperature-dependent viscous loss. Specifically, the TL-FNN predicts viscous loss using the instantaneous rotor speed and oil temperature as inputs, and the predicted loss is used in the temperature-aware model. The updated oil temperature is then fed back to the TL-FNN, allowing the viscous loss to be recalculated during transient evaluation.

IV. RESULTS AND VALIDATION

A. Case Study

An oil-flooded high-speed PM machine intended for an aerospace EHA is designed and prototyped to validate the proposed method, as illustrated in Figs. 1 and 11. The machine is designed for a maximum speed of 22 000 r/min to enhance power density. The machine specifications and design parameters are listed in Table I, and the material properties are listed in Table II. The rotor adopts a surface-mounted PM configuration featuring a quasi-Halbach array.

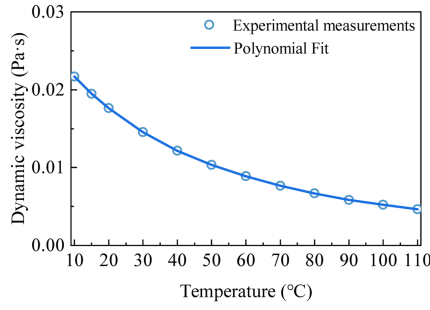


Fig. 7. Experimentally measured dynamic viscosity of the 15# hydraulic oil versus temperature.

TABLE III
Sampling Ranges and Levels for TL-Based Viscous Loss Modeling

Parameter	Range	Level (Analytical)	Level (CFD)
Rotational speed (r/min)	[3000, 27 000]	241	9
Temperature (°C)	[20, 100]	81	9

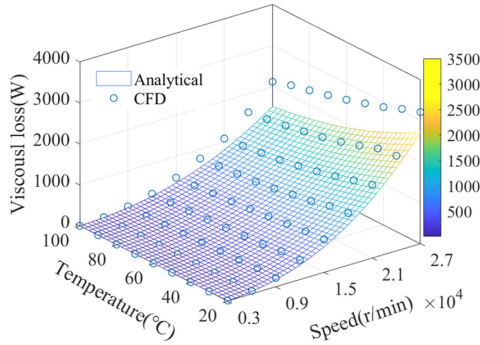


Fig. 8. Comparison of viscous losses predicted by the analytical and CFD models under different temperature and rotational speed conditions.

For TL-based viscous loss modeling, rotor speed and oil temperature are identified as the primary influencing factors. The experimentally measured dynamic viscosity of the oil is shown in Fig. 7. The viscosity at 20 °C is 0.0176 Pa·s, which is more than three times that at 100 °C (0.0052 Pa·s), demonstrating the strong temperature dependence of viscous losses. The sampling procedure is performed using the orthogonal experimental design (OED) method, with sampling ranges and levels summarized in Table III. Viscous loss data are computed from the analytical and CFD models to generate source and target domain training datasets, respectively.

As illustrated in Fig. 8, the analytical predictions agree well with CFD results at low speeds and temperatures. However, deviations increase under higher speed and higher temperature operating conditions, reaching a maximum error of 36.15%, as the analytical model cannot accurately capture turbulence effects. Although the analytical model can generate the complete source domain dataset within approximately 0.015 s and capture the overall viscous loss trend, its accuracy is insufficient for use as the final predictor, especially under complex-flow conditions. The CFD model used for generating the target-domain data employs a final mesh comprising approximately 120 million cells,

as depicted in Fig. 2. Simulations are carried out on a workstation with AMD EPYC 7452 processors (2.35 GHz, 64 cores, 256 GB RAM), with each case consuming approximately 14848 core-hours. Parallel computation is employed across multiple nodes to ensure timely completion within 24 h. Therefore, applying full CFD simulations over the entire speed–temperature range is impractical for repeated performance evaluation or design iteration due to the high computational cost.

B. TL-FNN Performance Evaluation

The complete analytical dataset ($n_s = 19\,521$) and a limited CFD dataset ($n_t = 9$) are employed as the source domain and target domain, respectively, for FNN pretraining and subsequent TL-FNN fine-tuning. All input features are normalized using z-score normalization. For hyperparameter settings, the number of neurons in each of the two hidden layers is set to $n_1 = n_2 = 50$, with the tanh function selected as the activation function. The learning rates are set to 0.01 for pretraining and 0.001 for fine-tuning.

To evaluate prediction accuracy and generalization performance, the total 81 CFD samples are used as the test set. The coefficient of determination (R^2) and the symmetric mean absolute percentage error (SMAPE) are adopted as evaluation metrics, given by

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (36)$$

$$\text{SMAPE} = \frac{1}{N} \sum_{i=1}^N \frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2} \times 100\% \quad (37)$$

where N is the number of test samples, y_i and $\bar{y}$ denote the true value of the test data and its mean, respectively, $\hat{y}_i$ and represents the predicted value. R^2 represents the goodness of fit of the model, with values closer to 1 indicating a better capture of data trends and overall global fitting performance. SMAPE addresses the average relative error between predicted and true values, reflecting the prediction accuracy of the model.

Fig. 9 presents the prediction results on the test set, highlighting the deviation between predicted and true values. Here, FNN refers to the conventional FNN model trained directly on the target domain CFD dataset [$n_t = 9$, level(CFD) = 3] without pretraining. It is obvious that TL-FNN outperforms both pretrained FNN and FNN in prediction accuracy. Fig. 9(b) and (c) further compares the deviations of predicted values from true values for TL-FNN and FNN, demonstrate that TL-FNN achieves superior prediction accuracy (SMAPE = 5.881%, $R^2 = 0.996$), validating the effectiveness of the proposed method. By leveraging low-cost analytical viscous loss data through model-based TL, the TL-FNN achieved high prediction accuracy using only nine high-fidelity CFD samples. Notably, the analytical dataset used for pretraining consists of 19 521 samples and is generated within 0.015 s, whereas a single CFD simulation requires approximately 24 h. This demonstrates the

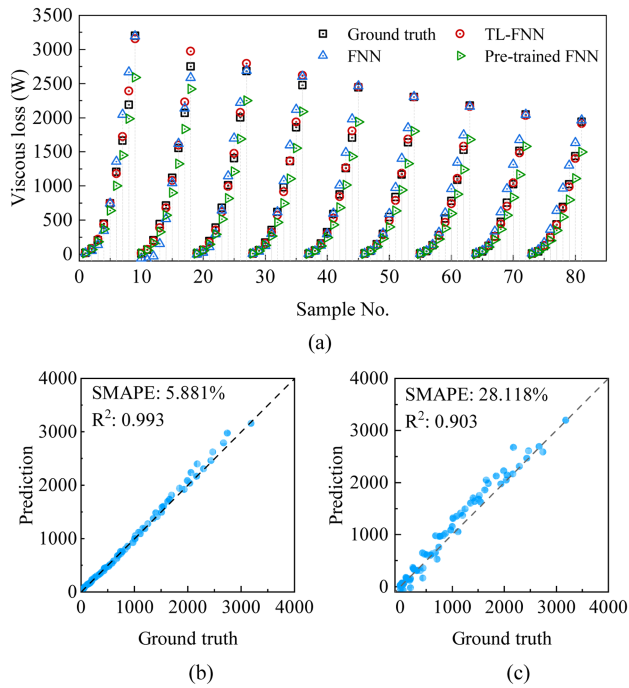


Fig. 9. Prediction performance comparison on the test set. (a) Viscous loss prediction results of different models. (b) Prediction performance of the TL-FNN. (c) Prediction performance FNN.

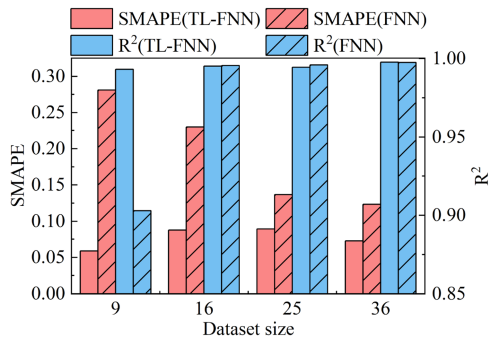


Fig. 10. Prediction performance comparison of different target domain CFD dataset sizes.

potential of the proposed TL-FNN in effectively mitigating the reliance on extensive high-fidelity data.

Fig. 10 examines the impact of varying target domain CFD dataset sizes n_t on TL-FNN's prediction accuracy and generalization, compared with FNN trained on the same samples. As n_t increases, TL-FNN maintains SMAPE below 9%, with R^2 improving from 0.993 to 0.997. For FNN, R^2 rises from 0.903 to 0.996 as from 9 to 16, but SMAPE decreases slowly from 28.118%, remaining higher than TL-FNN's values. This indicates that the conventional FNN can capture the overall viscous loss trend when more samples are available, but may suffer from systematic bias and larger errors in small-sample cases. In contrast, the proposed TL-FNN strategy improves generalization and mitigates overfitting with limited high-fidelity CFD samples through source domain parameter transfer and target domain fine-tuning.

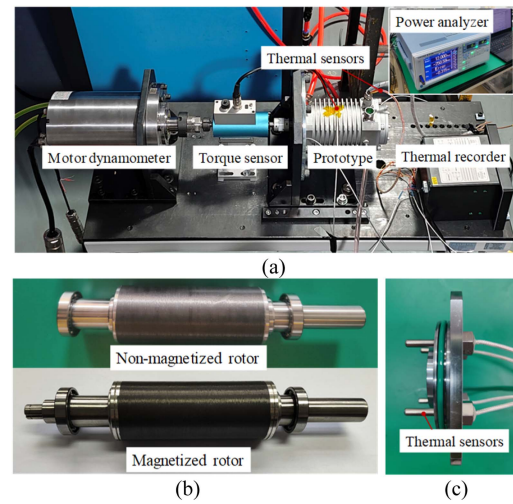


Fig. 11. Test rig and measurement setup. (a) Test rig with prototype assembly. (b) Nonmagnetized and magnetized rotors. (c) Custom-designed sensor-integrated end cap.

C. Experimental Validation

Experimental validation is conducted to assess the predictive accuracy of the TL-FNN model and the HMDD temperature-dependent viscous loss analysis approach in quantifying viscous losses behavior of the oil-flooded high-speed motor prototype. The test rig and measurement setup are illustrated in Fig. 11(a). Temperature measurements employ thermal sensors PT100, with three pre-embedded in the windings of each phase during manufacturing and one attached to the housing. In addition, four probe-type PT100 sensors are inserted near the airgap through a custom-designed sensor-integrated end cap, enabling more accurate measurement of the local fluid temperature, as shown in Fig. 11(c). The prototype is driven to steady-state speeds using the motor dynamometer, and the rotational speed, torque, and temperature are recorded.

Since no-load core loss and PM eddy-current loss still exist in a magnetized PM machine due to PM excitation, a nonmagnetized rotor with the same geometry is employed to isolate viscous losses by eliminating these electromagnetic loss components, as shown in Fig. 11(b). For a given machine geometry, the viscous loss is mainly governed by rotor speed and oil temperature-dependent viscosity, which determine the air-gap flow characteristics and are preserved in the test.

Fig. 12 presents a comparison between the TL-FNN predicted viscous losses and experimental measurements under various temperature and speed conditions, with the maximum rotational speed up to 27 000 r/min. The TL-FNN tends to slightly underestimate the viscous losses, particularly at elevated temperatures and speeds, yielding a SMAPE of 6.28%. This deviation primarily arises from the inherent prediction accuracy and necessary modeling simplifications of the of the CFD model, together with surrogate fitting error and experimental uncertainty. Nevertheless, the overall accuracy remains acceptable for engineering applications.

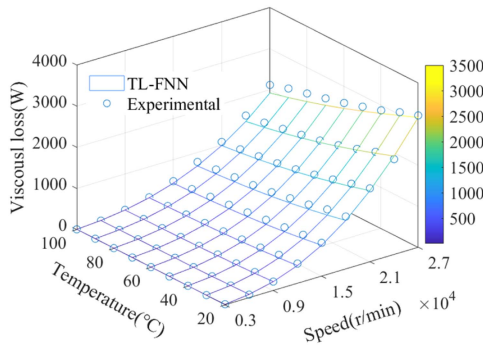


Fig. 12. Comparison of TL-FNN predicted and measured viscous losses under various temperature and rotational speed conditions.

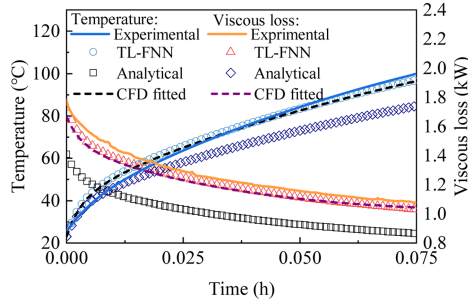


Fig. 13. Comparison of predicted and measured transient temperature and viscous loss at 21 000 r/min.

TABLE IV

Estimation Errors of Transient Temperature and Viscous Loss at Multiple Rotational Speeds

Performance	Indicator	Rotational speed (r/min)			
		6,000	12,000	18,000	24,000
Temperature	SMAPE	3.184%	4.420%	3.719%	5.205%
Viscous loss	SMAPE	1.540%	7.238 %	5.485%	7.753%

To validate the HMDD temperature-dependent viscous loss analysis approach, the coupled temperature-loss behavior is evaluated at 21 000 r/min, as shown in Fig. 13. When integrated with the TL-FNN, the HMDD framework achieves high prediction accuracy, with SMAPE values of 3.52% for temperature and 4.01% for viscous loss comparing to experimental results. This performance is comparable to that obtained using a viscous model directly fitted with CFD-predicted samples at the same speed. In contrast, when the analytical viscous loss model is employed, its tendency to underestimate viscous losses leads to progressively increasing temperature prediction errors over time, resulting in SMAPE values of 10.63% for temperature and 22.67% for viscous loss.

Further experimental validations at constant rotational speeds of 6000, 12 000, 18 000, and 24 000 r/min are presented in Fig. 14, while Table IV quantifies the estimation errors using SMAPE. The HMDD framework maintains high accuracy in coupled temperature-loss prediction, with the maximum SMAPE values of 5.21% for temperature and 7.75% for viscous loss. These findings validate the

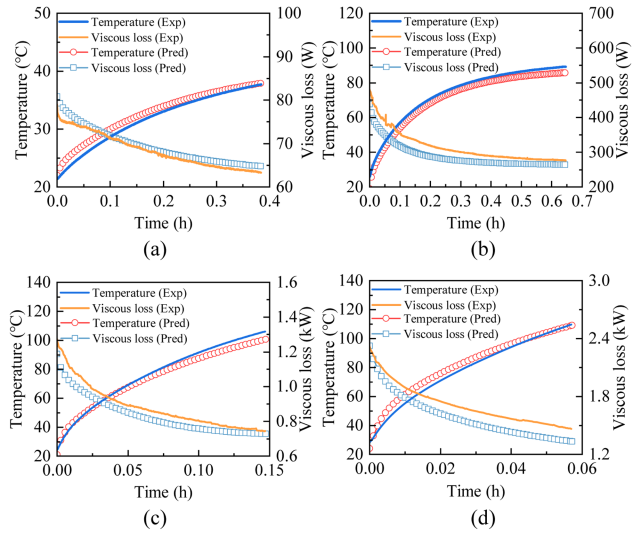


Fig. 14. Comparison of predicted and measured transient temperature and viscous loss at multiple rotational speeds. (a) 6 000 r/min. (b) 12 000 r/min. (c) 18 000 r/min. (d) 24 000 r/min.

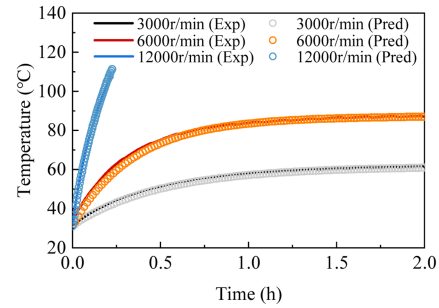


Fig. 15. Comparison of predicted and measured transient winding temperature at multiple rotational speeds of 3000 r/min, 6000 r/min, and 12 000 r/min.

feasibility and effectiveness of incorporating the TL-FNN model as a surrogate predictor in the proposed HMDD approach.

To further validate the proposed HMDD approach, temperature-rise tests are conducted on the prototype equipped with the magnetized rotor. The prototype is driven by an inverter using the $i_d = 0$ control strategy. Fig. 15 compares the predicted and experimental transient winding temperatures at 3000, 6000, and 12 000 r/min under a constant load of 1 N·m. The predictions closely match the experimental measurements, with SMAPE values of 3.59%, 3.39%, and 2.61%, respectively, confirming the model accuracy across different operating conditions. Fig. 16 presents the temperature rise curves of both the winding and housing at 22 000 r/min and 5.4 N·m, with maximum temperature discrepancies within 3 °C. These results indicate that accurate prediction of winding temperature indirectly reflects the ability of the HMDD model to capture the temperature-dependent viscous loss behavior and its impact on machine performance.

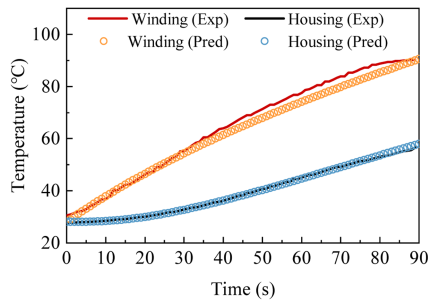


Fig. 16. Comparison of predicted and measured transient temperature at 22 000 r/min.

V. CONCLUSION

This article presents an HMDD approach for temperature-dependent viscous loss modeling in oil-flooded high-speed machines. By leveraging low-cost analytical viscous loss data through model-based TL, the TL-FNN is developed to construct an accurate viscous loss surrogate model, achieving high prediction accuracy (SMAPE = 5.88%, $R^2 = 0.996$) using only nine high-fidelity CFD samples selected through the OED method to cover the speed–temperature range, effectively addressing the challenge of data scarcity. A temperature-aware model based on a 3-D LPTN is incorporated to capture the temperature-loss interaction and enable performance evaluation under practical operating conditions, with an analytical temperature solution derived for efficient transient computation without compromising accuracy. Experimental validation over 3000–27 000 r/min and 20 °C–100 °C shows a maximum error of 7.8% in viscous loss prediction and 5.2% in oil temperature, while winding temperature deviations remain within 3 °C. These results demonstrate the capability of the HMDD model to capture temperature-dependent viscous loss behavior and its impact on machine performance.

Overall, the proposed HMDD framework provides an efficient and accurate solution for temperature-dependent viscous loss modeling and supports performance evaluation of oil-flooded high-speed machines for EHA applications. Nevertheless, the present modeling and validation are based on the given machine dimensions, oil type, and operating range, and additional CFD sampling or fine-tuning may be required when these conditions change significantly. Future work will focus on improving model generalization for variations in key machine dimensions and oil viscosity, as well as transient viscous loss prediction under rapid speed-varying operation.

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